

ISW Paper Revision Patch

Proposed Corrections to
“ISW Cross-Correlation Predictions for EFC with DESI DR1 Tracers”

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Companion to ISW Consistency Audit v1.1

Summary. The ISW Consistency Audit (v1.1) established that the published A_{ISW} values ($0.56 \rightarrow 0.15$) cannot be reproduced under any self-consistent channel—growth-driven, potential-driven, or covariance-weighted. This patch proposes minimal, defensible revisions to the ISW paper that preserve its core predictions while correcting the quantitative A_{ISW} claims.

1 What Must Change

Three categories of revision are needed, in order of severity.

1.1 Category 1: Retract specific A_{ISW} values

The published table reports A_{ISW} values per tracer (0.56, 0.43, 0.33, 0.29, 0.15, 0.20) as model predictions. A fully self-consistent computation yields $A_{\text{ISW}} \approx 0.87\text{--}0.93$ (growth-driven) with *no significant z -dependence*.

Proposed replacement:

Under the growth-driven interpretation (μ enters only the growth ODE, not the gravitational potential), the predicted ISW amplitude ratio is $A_{\text{ISW}} \approx 0.89 \pm 0.03$ across all DESI DR1 tracer bins, representing a uniform $\sim 11\%$ suppression relative to ΛCDM . The previously reported strong z -dependent suppression ($A_{\text{ISW}} = 0.56 \rightarrow 0.15$) is not reproduced under full ODE-consistent computation and has been withdrawn pending methodological review.

1.2 Category 2: Clarify the μ -channel interpretation

The original paper did not explicitly distinguish between:

- **Growth-driven:** $\Phi \propto D(a)/a$; μ affects ISW only through $D(a)$ and $f(a)$
- **Potential-driven (MG-type):** $\Phi \propto \mu(a) D(a)/a$; ISW kernel contains explicit $d\mu/d\ln a$ term

The Audit shows these give qualitatively different predictions: growth-driven yields positive $A_{\text{ISW}} \approx 0.89$; potential-driven yields *anti-correlation* ($A < 0$) for $z_{\text{eff}} \gtrsim 0.6$.

Proposed insertion (new subsection in Method):

The EFC gravitational coupling $\mu(a)$ enters the growth equation (1) and modifies $D(a)$ relative to Λ CDM. For ISW predictions, the critical question is whether μ also enters the Poisson equation for the metric potential Φ . In the growth-driven interpretation adopted here, $\Phi \propto D(a)/a$ and the ISW source is $\mathcal{S} \propto H(a)(D/a)(1-f)$, where $f = d \ln D / d \ln a$. The potential-driven alternative ($\Phi \propto \mu D/a$, yielding an additional $d\mu/d \ln a$ source term) produces anti-correlated ISW signal for tracers overlapping the transition epoch and is disfavoured by existing positive ISW detections.

1.3 Category 3: Revise the cancellation metric

The cancellation index $\mathcal{C}$ was defined as:

$$\mathcal{C} = 1 - \frac{|\sum_{\ell} C_{\ell}^{\text{tot}}|}{|\sum_{\ell} C_{\ell}^{K_{\mu}}| + |\sum_{\ell} C_{\ell}^{K_{\mu'}}|} \quad (1)$$

Under growth-driven ISW, there is no $K_{\mu'}$ term. The cancellation metric is undefined or identically zero.

Proposed replacement:

The cancellation index $\mathcal{C}$ quantifies the degree of cancellation between opposite-sign ISW source contributions. Under the growth-driven interpretation, $\mathcal{C}$ is not applicable (no separate $K_{\mu'}$ channel exists). The observable discriminator between EFC and Λ CDM in this case is the uniform suppression $A_{\text{ISW}} \approx 0.89$, detectable only through combined multi-tracer analysis with sufficient S/N .

2 What Can Be Preserved

The following elements of the original paper remain valid and should be retained.

Table 1: Elements of the ISW paper that survive the Audit.

Element	Status
Growth ODE, eq. (1)	Correct. $\mu(a)$ in growth equation is well-defined.
$\mu(a)$ parameterisation	Correct. $\beta = 0.16$, $a_t = 0.55$, $\sigma = 0.25$ are unchanged.
Limber approximation	Valid. Exact non-Limber differs by $< 3\%$.
Tracer definitions	Valid. DESI DR1 bins and bias values are unchanged.
$H(a)$ background	Valid. Fiducial Λ CDM background is correct.
Qualitative prediction	Valid. EFC predicts $A_{\text{ISW}} < 1$ for all tracers.
Falsification criterion	Valid (revised). If observed $A_{\text{ISW}} = 1.00 \pm 0.05$ across all bins, EFC growth-channel is excluded.

3 Revised Prediction Summary

Table 2: Revised EFC predictions for ISW cross-correlation. Growth-driven channel; $\ell \in [2, 60]$. “Original” column shown for reference (now withdrawn).

Tracer	z_{eff}	A_{ISW} (revised)	A_{ISW} (original)	$\mathcal{C}$
LRGz0	0.510	0.87	0.56	—
LRGz1	0.706	0.87	0.43	—
LRGz2	0.930	0.89	0.33	—
LRG+ELG	0.930	0.89	0.29	—
ELG	1.317	0.92	0.15	—
QSO	1.491	0.90	0.20	—

Key changes:

- A_{ISW} is nearly z -independent (0.87–0.92), not strongly declining.
- The suppression is $\sim 11\%$ relative to Λ CDM, not 44–85% as originally claimed.
- The cancellation metric $\mathcal{C}$ is not applicable under growth-driven ISW and is removed.

4 Impact on Validation Ledger

The Living Validation Ledger entry for ISW must be updated:

Table 3: Validation Ledger ISW entry: current vs. revised.

Field	Current (v1.6)	Revised
Status	Completed (no observational conflict)	Under revision
A_{ISW}	0.29–0.56 (bias-calibrated)	0.87–0.92 (growth-driven, self-consistent)
$\mathcal{C}$	0.59–0.96	Not applicable (growth-driven)
Falsification	$\mathcal{C} > 0.7$, $A < 0.5$ at z_t	$A_{\text{ISW}} = 1.00 \pm 0.05$ across all bins
Key discriminator	Tomographic $\mathcal{C}$ gradient	Uniform 11% suppression in multi-tracer stack

5 Recommended Wording for Updated Paper

5.1 Revised abstract (final paragraph)

Under self-consistent growth-driven computation, EFC predicts a uniform ISW amplitude suppression of $A_{\text{ISW}} \approx 0.89 \pm 0.03$ across all DESI DR1 tracer bins at $\ell \in [2, 60]$, representing an $\sim 11\%$ reduction relative to ΛCDM . This suppression is z -independent to within numerical precision and is consistent with all published ISW measurements within their current uncertainties. A definitive test requires a combined multi-tracer analysis with S/N sufficient to distinguish 11% amplitude shifts.

5.2 Revised falsification statement

If future tomographic ISW measurements with DESI spectroscopic bins yield $A_{\text{ISW}} = 1.00 \pm 0.05$ in a combined multi-tracer stack, the EFC growth-channel modification would be excluded at $> 2\sigma$. Conversely, observation of $A_{\text{ISW}} < 0.9$ across multiple bins would constitute positive evidence for the modified growth rate predicted by EFC.

6 Implementation Notes

- This patch applies to version DOI: [10.6084/m9.figshare.31301953](https://doi.org/10.6084/m9.figshare.31301953).
- The corrected pipeline code is available as `isw_channel_test.py` and `isw_amplitude_fit.py` in the Symbiose knowledge base.
- A revised Figshare upload should carry version suffix “v2—corrected” and reference the Audit note.
- The Validation Ledger should be updated to v1.7 with status “Under revision” until the revised paper is published.

End of revision patch. All corrections are traceable to ISW Consistency Audit v1.1.